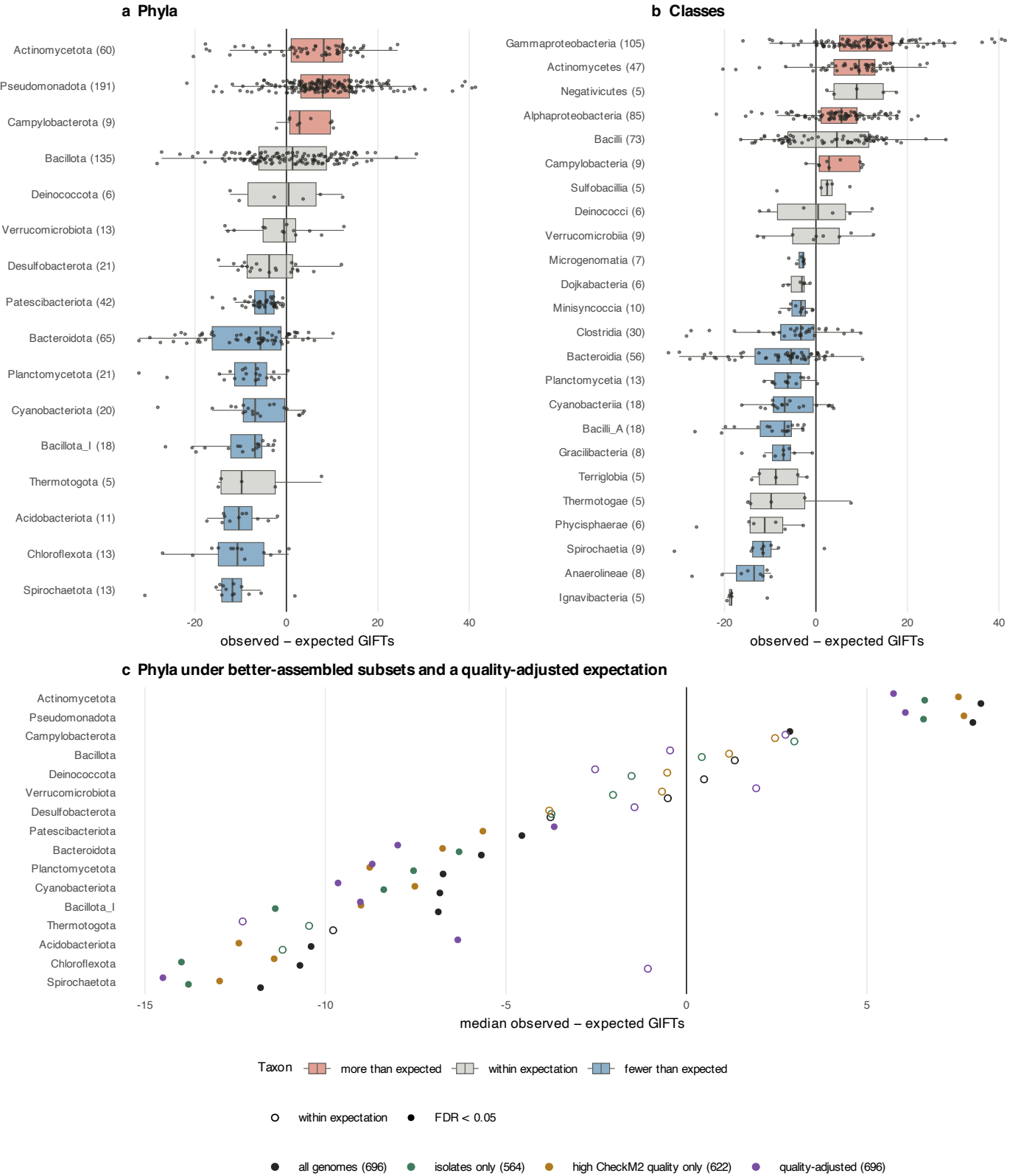


Taxa above and below the size expectation

Deviation of each genome from the Figure S9 expectation, for every phylum and class with at least 5 genomes (count in brackets). A box is coloured when the taxon's residuals differ from zero at a false discovery rate of 0.05.



Taxa are ordered by median deviation. The panel is phylogenetically enriched, not frequency weighted, and genomes within a taxon are not independent. In c, high CheckM2 quality is at least 95% complete and at most 5% contaminated, and the quality-adjusted expectation adds CheckM2 completeness and metagenome origin as covariates.